

Homework 2

Solutions

Exercise 2.1 (Hogg, Tanis and Zimmerman, Exercise 3.3-15). The "fill" problem is important in many industries, such as those making cereal, toothpaste, beer, and so on. If an industry claims that it is selling 12 ounces of its product in a container, it must have a mean greater than 12 ounces, or else the FDA will crack down, although the FDA will allow a very small percentage of the containers to have less than 12 ounces.

(a) If the content X of a container has a $\mathcal{N}(12.1, \sigma^2)$ distribution, find σ so that $\mathbb{P}(X < 12) = 0.01$.

(b) If $\sigma = 0.05$, find μ so that $\mathbb{P}(X < 12) = 0.01$.

Sol. This is a bit annoying to plug directly into a calculator, so I'm going to transform it a bit. If $X \sim \mathcal{N}(12.1, \sigma^2)$, then $\frac{X-12.1}{\sigma} \sim \mathcal{N}(0, 1)$. Now,

$$\mathbb{P}(X < 12) = \mathbb{P}(X - 12.1 < -0.1) = \mathbb{P}\left(\frac{X - 12.1}{\sigma} < \frac{-0.1}{\sigma}\right) = \Phi\left(-\frac{0.1}{\sigma}\right),$$

where Φ is the cumulative distribution function of the standard normal. Consulting a table for the standard normal, we see that

$$\Phi\left(-\frac{0.1}{\sigma}\right) = 0.01 \implies \frac{-0.1}{\sigma} \approx -2.33 \implies \boxed{\sigma \approx 0.04}.$$

On the other hand, if $\sigma = 0.05$, we have again that $\frac{X-\mu}{0.05} \sim \mathcal{N}(0, 1)$ and

$$\mathbb{P}(X < 12) = \mathbb{P}(X - \mu < 12 - \mu) = \mathbb{P}\left(\frac{X - \mu}{0.05} < \frac{12 - \mu}{0.05}\right) = \Phi\left(\frac{12 - \mu}{0.05}\right).$$

Consulting a table for the standard normal, we want again

$$\Phi\left(\frac{12 - \mu}{0.05}\right) = 0.01 \implies \frac{12 - \mu}{0.05} \approx -2.33 \implies \boxed{\mu \approx 12.1165}.$$

Exercise 2.2 (Panaretos, Exercise 7). Let X, Y be independent exponential random variables with rates λ_1 and λ_2 . Prove that $Z = \min(X, Y)$ is also exponential with rate $\lambda_1 + \lambda_2$.

Proof. Observe that

$$\mathbb{P}(Z > t) = \mathbb{P}(\min(X, Y) > t) = \mathbb{P}(X > t \text{ and } Y > t) = (1 - \mathbb{P}(X \leq t))(1 - \mathbb{P}(Y \leq t)) = (1 - F_X(t))(1 - F_Y(t)).$$

using independence. If $t \leq 0$ then, this product is one using the distribution function of $\text{Exp}(\lambda)$, which is given by

$$F_\lambda(x) := \begin{cases} 1 - e^{-\lambda x} & \text{if } x > 0 \\ 0 & \text{if } x \leq 0. \end{cases}$$

We also have then for $t > 0$ that

$$\mathbb{P}(Z > t) = e^{-\lambda_1 t} e^{-\lambda_2 t} = e^{-(\lambda_1 + \lambda_2)t} \implies F_Z(t) = \mathbb{P}(Z \leq t) = 1 - e^{-(\lambda_1 + \lambda_2)t}$$

and for $t \leq 0$ that $F_Z(t) = \mathbb{P}(Z \leq t) = 1 - 1 = 0$. Hence,

$$F_Z(t) := \begin{cases} 1 - e^{-(\lambda_1 + \lambda_2)t} & \text{if } t > 0 \\ 0 & \text{if } t \leq 0 \end{cases}$$

which uniquely characterizes $Z \sim \text{Exp}(\lambda_1 + \lambda_2)$. □

Exercise 2.3 (Zacks, Exercise 2.4.11). Let X and Y be independent standard normal random variables. What is the distribution of the distance of (X, Y) from the origin?

Sol. The distance to the origin is given by $d = \sqrt{X^2 + Y^2}$. $d^2 := X^2 + Y^2 \sim \chi^2(2)$, which has probability density function

$$f_{d^2}(x) = \frac{1}{2}e^{-\frac{x}{2}}.$$

We could find the density for d using the theorem we proved in class on Friday, but again I intend to show that just using the method is a bit easier to remember. The cumulative distribution function of this $\chi^2(2)$ random variable (which, by the way, is also the distribution of an $\text{Exp}(\frac{1}{2})$ random variable, is

$$F_{d^2}(x) = \begin{cases} 1 - e^{-x/2} & \text{if } x > 0 \\ 0 & \text{if } x \leq 0. \end{cases}$$

If $x > 0$, we compute

$$\mathbb{P}(d \leq x) = \mathbb{P}(d^2 \leq x^2) = F_{d^2}(x^2) = 1 - e^{-\frac{x^2}{2}}.$$

For $x \leq 0$, $F_d(x) = 0$. Differentiating, we find

$$f_d(x) = \begin{cases} xe^{-x^2/2} & \text{if } x > 0 \\ 0 & \text{if } x \leq 0 \end{cases}$$

which can be summarized as $f_d(x) := xe^{-x^2/2}\mathbf{1}_{x \geq 0}$. This is the definition of a $\chi(2)$ random variable, the square root of a $\chi^2(2)$ random variable.

Exercise 2.4 (Panaretos, Exercises 6 and 14). This exercise concerns the memoryless property of exponential random variables.

(a) Show that if $X \sim \text{Exp}(\lambda)$, then

$$\mathbb{P}(X \geq x + t | X \geq t) = \mathbb{P}(X \geq x), \quad x, t \geq 0. \quad (1)$$

Proof. We compute

$$\mathbb{P}(X \geq x + t | X \geq t) = \frac{\mathbb{P}(X \geq x + t \text{ and } X \geq t)}{\mathbb{P}(X \geq t)} = \frac{\mathbb{P}(X \geq x + t)}{\mathbb{P}(X \geq t)} = \frac{1 - F_X(x + t)}{1 - F_X(t)}$$

Using the formula for the cumulative distribution function given, for instance, in Problem 2, one finds

$$\mathbb{P}(X \geq x + t | X \geq t) = \frac{e^{-\lambda(x+t)}}{e^{-\lambda t}} = \frac{e^{-\lambda x} e^{-\lambda t}}{e^{-\lambda t}} = e^{-\lambda x} = 1 - F_X(x) = \mathbb{P}(X \geq x),$$

as desired. $\square$

(b) Suppose now that X is a continuous random variable with sample space $\{x \in \mathbb{R} : x \geq 0\}$. Suppose also that X satisfies the memoryless property (1), and that $\mathbb{P}(X > 0) > 0$. The goal of the next several exercises is to show that $X \sim \text{Exp}(\lambda)$ for some $\lambda \in (0, +\infty)$.

(i) Let $F_X(x)$ denote the distribution function of X , and $g(x) = 1 - F_X(x)$. Show that

$$g(x + y) = g(x)g(y).$$

(ii) Define $h(t) = -\ln g(t)$, and let $\lambda = h(1)$. Show that $h(k) = k\lambda$ for all $k \in \{1, 2, 3, \dots\}$.

(iii) Conclude that $h(t) = t\lambda$ for all rational $t > 0$.

(iv) What is the sign of λ ?

- (v) Since g is continuous by assumption, this implies that $h(t) = t\lambda$ for all real $t > 0$. Using $g(0) > 0$ and the continuity of g , show that $\lambda < +\infty$.
- (vi) What is the distribution function $F_X(x)$?
- (vii) Compute the probability density function of X , recognizing our exponential random variable.

Proof. We start with item (i). Observe that

$$g(u) = 1 - F(u) = \mathbb{P}(X > u) = \mathbb{P}(X \geq u) \quad (2)$$

where we have reinserted the equal sign in the inequality since the random variable is assumed to be continuous. We only consider $x, y \geq 0$ from this point onward, since the sample space of the random variable is $\{x \geq 0\}$. Now, the memoryless property tells us that

$$g(x) = \mathbb{P}(X \geq x) = \mathbb{P}(X \geq x + y | X \geq y) = \frac{\mathbb{P}(X \geq x + y \text{ and } x \geq y)}{\mathbb{P}(X \geq y)} = \frac{\mathbb{P}(X \geq x + y)}{\mathbb{P}(X \geq y)} = \frac{g(x + y)}{g(y)}$$

using (2). Rearranging, we find $g(x + y) = g(x)g(y)$.

Now, we will use this multiplicativity to show that $h(t) = -\ln g(t)$ has the requested property in (ii). First, we have

$$h(x + y) = -\ln g(x + y) = -\ln g(x)g(y) = -\ln g(x) - \ln g(y) = h(x) + h(y),$$

so that h is an additive function. $h(k) = k\lambda$ now follows by induction for all $k \in \mathbb{N}$; the base case is by definition, and if it is true for $k = n$ we have

$$h(n + 1) = h(n) + h(1) = \lambda n + \lambda = \lambda(n + 1).$$

So, $h(k) = k\lambda$ for all $k \in \{1, 2, 3, \dots\}$. This also implies $h(t) = t\lambda$ for all rational $t > 0$, as requested in (iii). Indeed, if $t = \frac{p}{q} > 0$, then

$$h(qt) = h(p) = p\lambda$$

but additivity also tells us that

$$h(qt) = h(t + t + \dots + t) = qh(t) \implies qh(t) = p\lambda \implies h(t) = \frac{p}{q}\lambda = t\lambda,$$

as desired. This establishes the first three points.

For the fourth point, observe that $g(1)$ is a probability, and thus lives between 0 and 1. Hence, $\ln g(1)$ is nonpositive, so $h(1)$ is nonnegative. Now, in point (5), one can use the definition of continuity to show that $h(t) = t\lambda$ for all $t > 0$; I don't ask you to prove this, this is largely an analysis exercise.

To see that λ is finite, it suffices to show that $g(1) > 0$, i.e. that $\mathbb{P}(X > 1) > 0$. Suppose for the sake of a contradiction that $g(1) = 0$ (it cannot be negative as a probability, so zero is the only other option); then, using (2),

$$0 = g(1) = g\left(\frac{1}{2} + \frac{1}{2}\right) = g\left(\frac{1}{2}\right)^2 \implies g\left(\frac{1}{2}\right) = 0.$$

Similarly,

$$0 = g\left(\frac{1}{2}\right) = g\left(\frac{1}{4} + \frac{1}{4}\right) = g\left(\frac{1}{4}\right)^2 \implies g\left(\frac{1}{4}\right) = 0.$$

Indeed, continuing inductively, one has $g\left(\frac{1}{2^n}\right) = 0$ for all $n \geq 1$; if we have $g\left(\frac{1}{2^k}\right) = 0$ then

$$0 = g\left(\frac{1}{2^k}\right) = g\left(\frac{1}{2^{k+1}} + \frac{1}{2^{k+1}}\right) = g\left(\frac{1}{2^{k+1}}\right)^2 \implies g\left(\frac{1}{2^{k+1}}\right) = 0.$$

which establishes the inductive step. Now, $\lim_{n \rightarrow \infty} \frac{1}{2^n} = 0$, so it follows that $g(0) = \lim_{n \rightarrow \infty} g\left(\frac{1}{2^n}\right) = \lim_{n \rightarrow \infty} 0 = 0$ by continuity. However, we assume in the problem statement that $g(0) > 0$, so this is a contradiction. Hence, $g(1) > 0$, and $\lambda < +\infty$, establishing point (v).

Rearranging $h(t) = t\lambda = -\ln g(t)$, we find $g(t) = e^{-\lambda t}$, and hence

$$F_X(x) = \begin{cases} 1 - e^{-\lambda x} & \text{if } x > 0 \\ 0 & \text{if } x \leq 0. \end{cases}$$

This implies that $f_X(x) = F'_X(x) = \lambda e^{-\lambda x} \mathbf{1}_{x>0}$, and that $X \sim \text{Exp}(\lambda)$. □